National University of Singapore

MA1522 - Linear Algebra for Computing

Final Exam

(AY2025/2026 Semester 1)

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS

1. This is a NON-SECURE BLOCK INTERNET assessment. You are allowed to refer to any soft copy or hard copy notes. All soft copy notes must be accessed through the device you are using for the exam.
2. This assessment contains 2 sections for a total of **80 marks**. You are required to answer ALL questions.
3. There are **15** questions in Section A, with **3 marks** per question. For each question, choose all the correct options.
4. Submit your answers for Section A in Examplify.
5. Section B consist of **4** (**FOUR**) questions, for a total of **55 marks**.
6. The questions for section B can be found in an attachment in question 16 of Examplify.
7. Write your answers for section B in the answer booklet provided.
8. Fill in your student number (e.g. A1234567Z) on the cover page of the answer booklet, in the space provided. **Do not write your name**.
9. **Begin each question on a new page** and indicate the question number clearly at the top of each page.
10. You are allowed to use MATLAB and any offline calculator. However, you should lay out systemically the various steps in the calculations for section B.
11. Hand in all the answer booklet at the end of the examination.
12. You are not allowed exit Examplify without informing the examiner. Type in the exit password in the blank provided in question 16 in Examplify.

Section A

1. (3 points) Let
$$\mathbf{A} = \begin{pmatrix} 2 & 2a-1 & -a-2 \\ 1 & a+1 & a-2 \\ 0 & -1 & 4-a \\ 1 & a & 2 \end{pmatrix}.$$
Which of the following statements are true? Select all that applies.
   (a) $\mathbf{A}^T\mathbf{A}$ is invertible any $a \in \mathbb{R}$.
   (b) There exists a $\mathbf{b} \in \mathbb{R}^4$ such that $\mathbf{Ax} = \mathbf{b}$ is inconsistent.
   (c) $\mathbf{A}$ is invertible.
   (d) $\text{rank}(\mathbf{A}) = 2$ for exactly 1 value of a.
   (e) We can find some values of a such that $\mathbf{A}$ has a right inverse.

2. (3 points) Let
$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & -1 & 3 \end{pmatrix}.$$
Which of the following statements are true? Select all that applies.
   (a) 1 is an eigenvalue of $\mathbf{A}$.
   (b) There exists an eigenvalue with algebraic multiplicity 2.
   (c) $\mathbf{A}$ is diagonalizable.
   (d) There exists an eigenvalue with geometric multiplicity 2.
   (e) There is a nonzero vector $\mathbf{v}$ such that $\mathbf{Av} = \mathbf{0}$.

3. (3 points) Let $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 6 & 3 \end{pmatrix}$. Compute its singular values. Select all that applies.
   (a) $\mathbf{A}$ has one non-zero singular value.
   (b) A singular value of $\mathbf{A}$ is $2\sqrt{15}$.
   (c) $\mathbf{A}$ has two non-zero singular values.
   (d) A singular value of $\mathbf{A}$ is $\sqrt{3}$.
   (e) $\mathbf{A}$ has no singular value decomposition.

4. (3 points) Let $\mathbf{A} = \mathbf{LU}$ be a LU-factorization of $\mathbf{A}$ with
$$\mathbf{U} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{pmatrix}.$$
Which of the following statements are true? Select all that applies.

Question 4 continues...

(a) $\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$ is a solution to the homogeneous system $\mathbf{Ax} = \mathbf{0}$.

(b) Rank of $\mathbf{A}$ is 2.

(c) We can find a $\mathbf{b} \in \mathbb{R}^3$ such that $\mathbf{Ax} = \mathbf{b}$ has a unique solution.

(d) We do not have enough information to determine $\det(\mathbf{A})$.

(e) The column space of $\mathbf{A}$ is spanned by $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 5 \\ 0 \end{pmatrix} \right\}$.

5. (3 points) Suppose $\mathbf{A} = \mathbf{PDP}^{-1}$, where $\mathbf{P} = (\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3)$ has linearly independent columns and

$$\mathbf{D} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Which of the following statements are true? Select all that apply.

(a) $2\mathbf{v}_2 - 9\mathbf{v}_3$ is an eigenvector of $\mathbf{A}$ associated to eigenvalue 2.

(b) The determinant of $\mathbf{A}$ is -4.

(c) The characteristic polynomial of $\mathbf{A}$ is $(x+1)(x-2)$.

(d) $5\mathbf{v}_1 + 3\mathbf{v}_2$ is an eigenvector of $\mathbf{A}$ associated to eigenvalue 2.

(e) The geometric multiplicity of eigenvalue 2 is 1.

6. (3 points) Which of the following subsets of $\mathbb{R}^3$ are subspaces? Select all that apply.

(a) $\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \,\middle|\, \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = 0 \right\}$

(b) $\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \,\middle|\, x = 0 \text{ and } y = 0 \right\}$

(c) $\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \,\middle|\, \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} y \\ z \\ x \end{pmatrix} = 0 \right\}$

(d) $\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \,\middle|\, x^2 = y^2 \right\}$

(e) $\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \,\middle|\, x = 0 \text{ or } y = 0 \right\}$

7. (3 points) Let $W = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \,\middle|\, x + y + z = 0 \right\}$. Which of the following sets form a basis for W? Select all that apply.

(a) $\left\{ \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}$

(b) $\left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right\}$

(c) $\left\{ \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix} \right\}$

(d) $\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}$

(e) $\left\{ \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\}$

8. (3 points) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 2 \end{pmatrix}.$$

Which of the following vectors is in the column space of $\mathbf{A}$? Select all that apply.

(a) $\begin{pmatrix} 3 \\ 6 \\ 3 \end{pmatrix}$

(b) $\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$

(c) $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

(d) $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

(e) $\begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$

9. (3 points) Let $\mathbf{A}$ be a rank 2 4×3 matrix and suppose $\mathbf{A}\begin{pmatrix}1\\2\\1\end{pmatrix} = \mathbf{0}$. It is known that $\begin{pmatrix}0\\1\\-1\end{pmatrix}$ is a least–squares solution of $\mathbf{Ax} = \mathbf{b}$. Which of the following vectors are also least–squares solutions to $\mathbf{Ax} = \mathbf{b}$? Select all that apply.

   (a) $\begin{pmatrix}1\\3\\0\end{pmatrix}$

   (b) $\begin{pmatrix}-2\\-3\\-3\end{pmatrix}$

   (c) $\begin{pmatrix}0\\0\\0\end{pmatrix}$

   (d) $\begin{pmatrix}1\\2\\1\end{pmatrix}$

   (e) $\begin{pmatrix}1\\1\\-1\end{pmatrix}$

10. (3 points) Every morning, Rachel chooses one of three cafés near her office for breakfast: Latte Love (L), Brew Bros (B), or Espresso Express (E). She never visits the same café two mornings in a row. If she visits Latte Love one morning, there is a 30% chance she will go to Brew Bros next. If she visits Brew Bros, she will go to Latte Love next with probability 60%. If she visits Espresso Express, there is a 50% chance she will go to Latte Love next.

    (a) The probability that Rachel visits Latte Love again two days after visiting it today is 0.53.

    (b) In the long run, Rachel will spend approximately 0.36 of the time in Espresso Express.

    (c) In the long run, Rachel spends equal proportions of time at all three cafés.

    (d) The probability that Rachel visits Espresso Express two days after visiting Latte Love today is approximately 0.37.

    (e) The probability that Rachel visits Latte Love three days after visiting Brew Bros is 0.12.

National University of Singapore

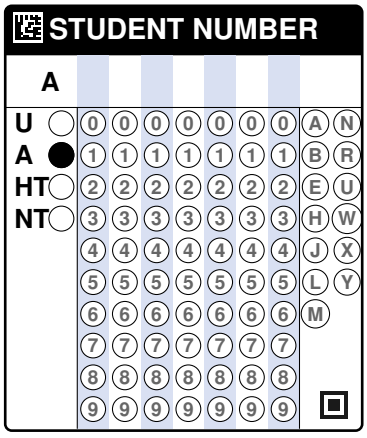
MA1522 - Linear Algebra for Computing

Final Exam

(AY2025/2026 Semester 1)

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS



1. Section B consistent of **4** (**FOUR**) questions, for a total of **50 marks**.
2. Write your answers for section B in the answer booklet provided.
3. Fill in and shade your student number (e.g. A1234567Z) on the cover page of the answer booklet, in the space provided. **Do not write your name**.
4. There are a total of **13** printed page, including the cover page, and 4 extra pages at the end.
5. Hand in all the answer booklet at the end of the examination.
6. You are not allowed to exit Examplify without informing the examiner. Type in the exit password in the blank provided in question 11 in Examplify.

1. Let

$$V = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \middle| \; 2x_1 - 3x_2 - 2x_3 + 3x_4 = 0 \right\}$$

and

$$T = \left\{ \mathbf{u}_1 = \begin{pmatrix} 1 \\ -a \\ 1 \\ -2 \end{pmatrix}, \mathbf{u}_2 = \begin{pmatrix} a \\ 2 \\ -1 \\ 0 \end{pmatrix}, \mathbf{u}_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

(a) (4 points) For what value(s) of a is T a basis for V? Justify your answer.

(b) (5 points) Assume the condition(s) on a from part (a) holds so that T is a basis for V. Determine which of the following vectors lie in V. For those that lie in V, write their coordinates relative to the basis T. For those that do not lie in V, find their orthogonal projection onto V:

$$\mathbf{w}_1 = \begin{pmatrix} -1 \\ 1 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} -2 \\ 3 \\ 1 \\ -1 \end{pmatrix}, \quad \mathbf{w}_3 = \begin{pmatrix} 1 \\ 2 \\ -2 \\ 0 \end{pmatrix}.$$

(c) (3 points) Assume the condition(s) on a from part (a) holds so that T is a basis for V. Let

$$S = \left\{ \mathbf{v}_1 = \begin{pmatrix} -2 \\ 0 \\ 1 \\ 2 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 3 \\ 0 \\ 0 \\ -2 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} -3 \\ 1 \\ 0 \\ 3 \end{pmatrix} \right\}.$$

Find the transition matrix from the basis T to the basis S.

2. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & a \\ 1 & 1 & 2 & 1 \end{pmatrix}, \quad a \in \mathbb{R}.$$

(a) (3 points) For different values of $a \in \mathbb{R}$, determine the rank of $\mathbf{A}$. For which value(s) of a does $\mathbf{A}$ have full rank? Explain how you derive your answer.

(b) (5 points) For the values of a such that $\mathbf{A}$ has full rank, determine whether $\mathbf{A}$ admits a left inverse, a right inverse, both, or neither. Then find all such inverses, expressing your answer in terms of a.

(c) (2 points) Find a basis for the column space of $\mathbf{A}$ when $\mathbf{A}$ is not of full rank.

3. Let $\mathbf{A}$ be a 5×4 real matrix with singular value decomposition

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T,$$

where

$$\mathbf{U} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 & 0 & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{\Sigma} = \begin{pmatrix} \sqrt{3} & 0 & 0 & 0 \\ 0 & \sqrt{3} & 0 & 0 \\ 0 & 0 & \sqrt{2} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The first three columns of $\mathbf{V}$ are given as

$$\mathbf{v}_1 = \begin{pmatrix} 0 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ 0 \\ 0 \\ \frac{\sqrt{2}}{2} \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

(a) (3 points) Find the rank of $\mathbf{A}$? Explain how you derive your answer.

(b) (2 points) Write down the eigenvalues of $\mathbf{A}^T\mathbf{A}$.

(c) (4 points) Let $\mathbf{w} = \begin{pmatrix} 1 \\ 2 \\ 1 \\ -1 \\ 1 \end{pmatrix}$. Find the projection of $\mathbf{w}$ onto the column space of $\mathbf{A}$. Explain how you derive your answer.

(d) (3 points) Find the general solution to the homogeneous system $\mathbf{Ax} = \mathbf{0}$. Explain how you derive your answer.

(e) (2 points) Find the matrix $\mathbf{A}$. Explain how you derive your answer.

4. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear transformation defined by

$$T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} ax + by + d \\ bx + ay + dz \\ -cx + cy + \dfrac{1}{2}z \end{pmatrix},$$

where $a, b, c, d \in \mathbb{R}$ are constants.

(a) (4 points) Determine the conditions on a, b, c, d such that the kernel of T is non-trivial. Explain your reasoning clearly.

(b) (3 points) For all subsequent parts, assume that

$$T\left(\begin{pmatrix}6\\0\\1\end{pmatrix}\right) = \begin{pmatrix}4\\-2\\-1\end{pmatrix}.$$

Under this condition, find the standard matrix of T, and explain how you obtained your result.

(c) (4 points) Continuing to assume the condition in part (b) determine whether it is possible to find a basis $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ of $\mathbb{R}^3$ such that, for all $c_1, c_2, c_3 \in \mathbb{R}$,

$$T(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3) = c_1\mathbf{u}_1 + \frac{1}{2}c_2\mathbf{u}_2 + \frac{1}{3}c_3\mathbf{u}_3.$$

Justify your answer.

(d) (3 points) Continuing to assume the condition in part (b), define

$$T^2(\mathbf{x}) = T(T(\mathbf{x})),$$

and for every integer $n \geq 1$,

$$T^{n+1}(\mathbf{x}) = T(T^n(\mathbf{x})) \quad \text{for all } \mathbf{x} \in \mathbb{R}^3.$$

Find a nonzero vector $\mathbf{v} \in \mathbb{R}^3$ such that, for every $\mathbf{x} \in \mathbb{R}^3$ there is a $\alpha \in \mathbb{R}$ (depending on $\mathbf{x}$), with exactly one of the following limits holding:

$$\lim_{n\to\infty} T^n(\mathbf{x}) = \alpha\mathbf{v} \quad \text{or} \quad \lim_{n\to\infty} T^n(\mathbf{x}) = \mathbf{0}.$$